



llama-3.3-70b-instruct-fp8-fast

Text Generation • Meta

@cf/meta/llama-3.3-70b-instruct-fp8-fast

Llama 3.3 70B quantized to fp8 precision, optimized to be faster.

Model Info

Context 24,000 tokens

Window [↗](#)

Terms and [link ↗](#)
License

Model Info

Function calling ↗	Yes
------------------------------------	-----

Batch	Yes
-------	-----

Unit Pricing	\$0.29 per M input tokens, \$2.25 per M output tokens
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Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/meta/llama-3.3-70b-instruct-  
fp8-fast", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

✓ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/meta/llama-3.3-70b-instruct-  
fp8-fast", { messages });  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```


▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/llama-3.3-70b-instruct-fp8-fast",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/  
3.3-70b-instruct-fp8-fast \\\n-X POST \\\n-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \\\n-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, \\\n  { "role": "user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **lora** **string**

Name of the LoRA (Low-Rank Adaptation) model to fine-tune the base model.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **repetition_penalty** **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

- ▶ **1** **object**

- **messages** **array** required

An array of message objects representing the conversation history.

- **items** **object**

- **role** **string** required

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- **content** **string** required

The content of the message as a string.

- **functions** **array**

- **items** **object**

- **name** **string** required

- **code** **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- 1 **object**
 - **type** **string** required
Specifies the type of tool (e.g., 'function').
 - **function** **object** required
Details of the function tool.
 - **name** **string** required
The name of the function.
 - **description** **string** required
A brief description of what the function does.
 - **parameters** **object** required
Schema defining the parameters accepted by the function.
 - **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **response_format** **object**

- **type** **string**

- **json_schema**

- `raw` **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- `stream` **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

- 2 **object**

- `requests` **array**

- **items** **object**

- **external_reference** **string**

User-supplied reference. This field will be present in the response as well it can be used to reference the request and response. It's NOT validated to be unique.

- **prompt** **string** min 1

Prompt for the text generation model

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **top_p** **number** min 0 max 2

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Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- `response_format` **object**
 - `type` **string**
 - `json_schema`

Output

- ▶ `0` **object**
 - `response` **string** required
The generated text response from the model
 - `usage` **object**
Usage statistics for the inference request
 - `prompt_tokens` **number** 0
Total number of tokens in input
 - `completion_tokens` **number** 0
Total number of tokens in output
 - `total_tokens` **number** 0
Total number of input and output tokens

- **tool_calls** **array**

An array of tool calls requests made during the response generation

- **items** **object**

- **arguments** **object**

The arguments passed to be passed to the tool call request

- **name** **string**

The name of the tool to be called

- ▶ **1** **string**

- ▶ **2** **object**

- **request_id** **string**

The async request id that can be used to obtain the results.

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{
  "type": "object",
  "oneOf": [
    {
      "title":
"Meta_Llama_3_3_70B_Instruct_Fp8_Fast_Prompt",
      "properties": {
        "prompt": {
          "type": "string",
          "minLength": 1,
          "description": "The input
text prompt for the model to generate a
response."
        },
        "lora": {
          "type": "string",
          "description": "Name of the
LoRA (Low-Rank Adaptation) model to fine-tune
the base model."
        },
        "response_format": {
          "title": "JSON Mode",
          "type": "object",
          "properties": {
```

```
      "type": {
        "type": "string",
        "enum": [
          "json_object",
          "json_schema"
        ]
      },
      "json_schema": {}
    },
    "raw": {
      "type": "boolean",
      "default": false,
      "description": "If true, a
chat template is not applied and you must adhere
to the specific model's expected formatting."
    },
    "stream": {
      "type": "boolean",
      "default": false,
      "description": "If true,
the response will be streamed back incrementally
using SSE, Server Sent Events."
    },
    "max_tokens": {
```

```
      "type": "integer",
      "default": 256,
      "description": "The maximum
number of tokens to generate in the response."
    },
    "temperature": {
      "type": "number",
      "default": 0.6,
      "minimum": 0,
      "maximum": 5,
      "description": "Controls
the randomness of the output; higher values
produce more random results."
    },
    "top_p": {
      "type": "number",
      "minimum": 0.001,
      "maximum": 1,
      "description": "Adjusts the
creativity of the AI's responses by controlling
how many possible words it considers. Lower
values make outputs more predictable; higher
values allow for more varied and creative
responses."
    },
  },
```

```
"top_k": {  
  "type": "integer",  
  "minimum": 1,  
  "maximum": 50,  
  "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more focused;  
higher values introduce more variety and  
potential surprises."  
},  
"seed": {  
  "type": "integer",  
  "minimum": 1,  
  "maximum": 9999999999,  
  "description": "Random seed  
for reproducibility of the generation."  
},  
"repetition_penalty": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Penalty for  
repeated tokens; higher values discourage  
repetition."  
},
```

```
    "frequency_penalty": {
      "type": "number",
      "minimum": -2,
      "maximum": 2,
      "description": "Decreases
the likelihood of the model repeating the same
lines verbatim."
    },
    "presence_penalty": {
      "type": "number",
      "minimum": -2,
      "maximum": 2,
      "description": "Increases
the likelihood of the model introducing new
topics."
    }
  },
  "required": [
    "prompt"
  ]
},
{
  "title":
"Meta_Llama_3_3_70B_Instruct_Fp8_Fast_Messages",
  "properties": {
```

```
"messages": {  
  "type": "array",  
  "description": "An array of  
message objects representing the conversation  
history.",  
  "items": {  
    "type": "object",  
    "properties": {  
      "role": {  
        "type":  
  
"string",  
        "description":  
"The role of the message sender (e.g., 'user',  
'assistant', 'system', 'tool')."  
      },  
      "content": {  
        "type":  
  
"string",  
        "description":  
"The content of the message as a string."  
      }  
    },  
    "required": [  
      "role",  
      "content"
```

```
    ]
  }
},
"functions": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "name": {
        "type": "string"
      },
      "code": {
        "type": "string"
      }
    },
    "required": [
      "name",
      "code"
    ]
  },
  "tools": {
    "type": "array",
    "description": "A list of
tools available for the assistant to use.",
```



```
      "items": {
        "type": "object",
        "oneOf": [
          {
            "properties": {
              "name": {
                "type":
"string",

"description": "The name of the tool. More
descriptive the better."
              },

"description": {
                "type":
"string",

"description": "A brief description of what the
tool does."
              },

"parameters": {
                "type":
"object",
```

```
"description": "Schema defining the parameters  
accepted by the tool.",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```

```
"description": "The type of the parameters  
object (usually 'object')."
```

```
},
```

```
"required": {
```

```
"type": "array",
```

```
"description": "List of required parameter  
names.",
```

```
"items": {
```

```
"type": "string"
```

```
}  
  
    },  
  
    "properties": {  
  
        "type": "object",  
  
        "description": "Definitions of each parameter.",  
  
        "additionalProperties": {  
  
            "type": "object",  
  
            "properties": {  
  
                "type": {  
  
                    "type": "string",  
  
                    "description": "The data type of the parameter."  
  
                },  
  
                "description": {
```

```
"type": "string",

"description": "A description of the expected
parameter."

},

"required": [

"type",

"description"

]

},

"required": [

"type",
```

"properties"

]

}

},

"required": [

"name",

"description",

"parameters"

]

},

{

"properties": {

"type": {

"type":

"string",

"description": "Specifies the type of tool
(e.g., 'function')."

},

"function":

{

"type":

"object",

```
"description": "Details of the function tool.",

"properties": {

  "name": {

    "type": "string",

    "description": "The name of the function."
  },

  "description": {

    "type": "string",

    "description": "A brief description of what the
function does."
  },

  "parameters": {

    "type": "object",

    "description": "Schema defining the parameters
```

```
accepted by the function.",

"properties": {

"type": {

"type": "string",

"description": "The type of the parameters
object (usually 'object')."

},

"required": {

"type": "array",

"description": "List of required parameter
names.",

"items": {

"type": "string"

}

}
```

```
},  
  
"properties": {  
  
  "type": "object",  
  
  "description": "Definitions of each parameter.",  
  
  "additionalProperties": {  
  
    "type": "object",  
  
    "properties": {  
  
      "type": {  
  
        "type": "string",  
  
        "description": "The data type of the parameter."  
  
      },  
  
      "description": {
```



```
"type": "string",

"description": "A description of the expected
parameter."

}

},

"required": [

"type",

"description"

]

}

}

},

"required": [
```

```
"type",

"properties"

]

    }

    },

"required": [

"name",

"description",

"parameters"

]

    }

    },

    "required": [

        "type",

        "function"

    ]

    }

]
```

```
    }  
  },  
  "response_format": {  
    "title": "JSON Mode",  
    "type": "object",  
    "properties": {  
      "type": {  
        "type": "string",  
        "enum": [  
          "json_object",  
          "json_schema"  
        ]  
      },  
      "json_schema": {}  
    }  
  },  
  "raw": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true, a  
chat template is not applied and you must adhere  
to the specific model's expected formatting."  
  },  
  "stream": {  
    "type": "boolean",
```

```
        "default": false,
        "description": "If true,
the response will be streamed back incrementally
using SSE, Server Sent Events."
    },
    "max_tokens": {
        "type": "integer",
        "default": 256,
        "description": "The maximum
number of tokens to generate in the response."
    },
    "temperature": {
        "type": "number",
        "default": 0.6,
        "minimum": 0,
        "maximum": 5,
        "description": "Controls
the randomness of the output; higher values
produce more random results."
    },
    "top_p": {
        "type": "number",
        "minimum": 0.001,
        "maximum": 1,
```

`"description": "Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses."`

`},`

`"top_k": {`

`"type": "integer",`

`"minimum": 1,`

`"maximum": 50,`

`"description": "Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises."`

`},`

`"seed": {`

`"type": "integer",`

`"minimum": 1,`

`"maximum": 9999999999,`

`"description": "Random seed for reproducibility of the generation."`

`},`

`"repetition_penalty": {`

```
        "type": "number",
        "minimum": 0,
        "maximum": 2,
        "description": "Penalty for
repeated tokens; higher values discourage
repetition."
    },
    "frequency_penalty": {
        "type": "number",
        "minimum": -2,
        "maximum": 2,
        "description": "Decreases
the likelihood of the model repeating the same
lines verbatim."
    },
    "presence_penalty": {
        "type": "number",
        "minimum": -2,
        "maximum": 2,
        "description": "Increases
the likelihood of the model introducing new
topics."
    }
},
"required": [
```

```
      "messages"
    ]
  },
  {
    "title": "Async Batch",
    "type": "object",
    "properties": {
      "requests": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {

"external_reference": {
          "type":
"string",
          "description":
"User-supplied reference. This field will be
present in the response as well it can be used
to reference the request and response. It's NOT
validated to be unique."
        },
        "prompt": {
          "type":
"string",
```

```
        "minLength": 1,
        "description":
"Prompt for the text generation model"
    },
    "stream": {
        "type":
"boolean",
        "default":
false,
        "description":
"If true, the response will be streamed back
incrementally using SSE, Server Sent Events."
    },
    "max_tokens": {
        "type":
"integer",
        "default": 256,
        "description":
"The maximum number of tokens to generate in the
response."
    },
    "temperature": {
        "type":
"number",
        "default": 0.6,
```



```
        "minimum": 0,
        "maximum": 5,
        "description":
"Controls the randomness of the output; higher
values produce more random results."
    },
    "top_p": {
        "type":
"number",
        "minimum": 0,
        "maximum": 2,
        "description":
"Adjusts the creativity of the AI's responses by
controlling how many possible words it
considers. Lower values make outputs more
predictable; higher values allow for more varied
and creative responses."
    },
    "seed": {
        "type":
"integer",
        "minimum": 1,
        "maximum":
9999999999,
```

```
        "description":  
        "Random seed for reproducibility of the  
        generation."  
    },  
  
    "repetition_penalty": {  
        "type":  
        "number",  
        "minimum": 0,  
        "maximum": 2,  
        "description":  
        "Penalty for repeated tokens; higher values  
        discourage repetition."  
    },  
    "frequency_penalty":  
    {  
        "type":  
        "number",  
        "minimum": 0,  
        "maximum": 2,  
        "description":  
        "Decreases the likelihood of the model repeating  
        the same lines verbatim."  
    },
```

```
        "presence_penalty":  
  
        {  
  
            "type":  
  
            "number",  
  
            "minimum": 0,  
            "maximum": 2,  
            "description":  
            "Increases the likelihood of the model  
            introducing new topics."  
        },  
        "response_format":  
  
        {  
  
            "title": "JSON  
            Mode",  
  
            "type":  
  
            "object",  
  
            "properties": {  
                "type": {  
                    "type":  
  
                    "string",  
  
                    "enum":  
  
                    [  
  
                    "json_object",
```

```
"json_schema"
```

```
]
```

```
},
```

```
"json_schema": {}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
]
```

```
}
```

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